

PRACTICE EXERCISES



Module 2: SQL Fundamentals (DDL & DML)

Module 2 - 8 Hours

SQL Fundamentals (DDL & DML)

Instructions

Work against your own `icicibank` database, which must already contain the nine core tables plus the `fixed_deposit` table with 20 pilot bookings.

Level	Meaning	Expected time
Level 1	Recall and understanding. Answer in your own words.	30 min
Level 2	Apply. Write and run SQL.	60 min
Level 3	Design and judgement. Justify your answer.	30 min

Rules:

- Every SQL answer must include the statement **and** the output you got.
- Where a statement is meant to fail, record the exact error message.
- Wrap anything destructive in `BEGIN; ... ROLLBACK;`.

Do not leave the database modified. Modules 3 onward expect 30 customers, 36 accounts and 20 fixed deposits.

How much to attempt

The questions marked **(Core)** are the required set. They fit the time above, and together they cover every objective of the module. Everything unmarked is **stretch work**: attempt it if you finish early, or take it home. **You are not behind if you have completed only the core set.**

Level	Core set	Stretch
Level 1	Q1, Q2, Q4, Q5, Q7, Q9, Q10	Q3, Q6, Q8
Level 2	Q11, Q12, Q17, Q19, Q20, Q24	Q13, Q14, Q15, Q16, Q18, Q21, Q22, Q23, Q25
Level 3	Q27, Q29	Q26, Q28

Level 1 - Understanding

Q1. (Core) For each column below, state the data type you would declare and give a one-line reason.

1. A customer's PAN number, always exactly 10 characters.
2. The interest amount credited on a fixed deposit at maturity.
3. The number of days a deposit runs, maximum 3,650.
4. The exact moment a UPI payment was authorised.
5. The date a loan instalment falls due.
6. A free-text branch note about a customer complaint.
7. A running counter of every transaction the bank has ever posted.
8. Whether a deposit auto-renews.

Q2. (Core) A developer proposes `interest_amount` `FLOAT` because "it handles decimals". Write the two-line SQL demonstration you would show them, and state in one sentence what the business impact is.

Q3. Complete the table.

Constraint	What it guarantees	One ICICI Bank example
NOT NULL	?	?
UNIQUE	?	?
PRIMARY KEY	?	?
FOREIGN KEY	?	?
CHECK	?	?
DEFAULT	?	?

Q4. (Core) State two differences between a `PRIMARY KEY` and a `UNIQUE` constraint, and explain why `deposit_id` was chosen as the primary key rather than `receipt_number`.

Q5. (Core) Explain the difference between `DELETE`, `TRUNCATE` and `DROP` across five dimensions: what is removed, whether `WHERE` is allowed, whether it can be rolled back, whether triggers fire, and speed on a large table.

Q6. ICICI Bank must keep seven years of transaction history for the regulator. A developer writes a monthly job that runs `DELETE FROM account_txn WHERE txn_date < CURRENT_DATE - 90`. State what is wrong and what the bank should do instead.

Q7. (Core) Predict the output of each, then run them.

```
SELECT NULL = NULL;
SELECT NULL <> NULL;
SELECT 5000 + NULL;
SELECT COALESCE(NULL, NULL, 'FALLBACK');
SELECT 'A' || NULL || 'B';
```

Q8. These two queries return different row counts. Explain why, in terms of operator precedence.

```
SELECT COUNT(*) FROM fixed_deposit
WHERE branch_id = 1 OR branch_id = 2 AND principal_amount > 500000;

SELECT COUNT(*) FROM fixed_deposit
WHERE (branch_id = 1 OR branch_id = 2) AND principal_amount > 500000;
```

Q9. (Core) A quarterly report filters with `WHERE txn_date BETWEEN DATE '2026-01-01' AND DATE '2026-03-31'`. It has been under-reporting for two years. Explain the defect and write the corrected predicate.

Q10. (Core) Why can a column alias defined in the `SELECT` list be used in `ORDER BY` but not in `WHERE`? Answer in terms of clause evaluation order.

Level 2 - Applying SQL

Build

Q11. (Core) ICICI Bank is launching a recurring deposit product. Create a table `recurring_deposit` with these requirements. Name every constraint.

Requirement	
<code>rd_id</code>	Whole number, primary key
<code>customer_id</code>	Must be an existing customer
<code>branch_id</code>	Must be an existing branch
<code>rd_number</code>	Up to 20 characters, must never repeat
<code>monthly_amount</code>	Money, at least 500
<code>interest_rate</code>	Percentage, above 0 and at most 12
<code>tenure_months</code>	Between 6 and 120
<code>start_date</code>	Defaults to today, mandatory
<code>maturity_date</code>	Mandatory, must be after <code>start_date</code>
<code>installments_paid</code>	Whole number, defaults to 0, cannot be negative
<code>status</code>	One of ACTIVE, MATURED, DEFAULTED, CLOSED, defaults to ACTIVE

Q12. (Core) Write and run one statement for each of these, and record the error message. All six must be rejected.

1. An RD with `monthly_amount` of 100.
2. An RD for `customer_id` 5555.
3. An RD whose `maturity_date` is before its `start_date`.
4. Two RDs with the same `rd_number`.
5. An RD with `status` of `PAUSED`.
6. An RD with `installments_paid` of -3.

Q13. Insert these three recurring deposits in a single statement.

<code>rd_id</code>	<code>customer_id</code>	<code>branch_id</code>	<code>rd_number</code>	<code>monthl y</code>	<code>rate</code>	<code>tenure</code>	<code>start</code>	<code>maturity</code>
8001	1001	1	RD2026000001	5000	7.10	24	2026-03-01	2028-03-01
8002	1005	5	RD2026000002	2500	6.85	12	2026-03-05	2027-03-05
8003	1012	4	RD2026000003	10000	7.35	60	2026-03-10	2031-03-10

Q14. The compliance team asks for the RD table to record a nominee. Add a nullable `nominee_name` column of up to 60 characters, then set the nominee for RD 8001 to 'Meera Reddy'.

Q15. Product decides the minimum monthly instalment rises to 1,000. Replace the existing check constraint. Before you do, write the query that tells you whether any existing row would violate the new rule.

Q16. Try to add a mandatory `scheme_code VARCHAR(8) NOT NULL` column to the populated `recurring_deposit` table. Record the error, then make it succeed.

Query

Q17. (Core) List every fixed deposit of 500,000 or more, showing receipt number, principal, rate and maturity date, largest first.

Q18. List fixed deposits booked between 1 and 10 March 2026 inclusive, sorted by booked date then principal descending.

Q19. (Core) Show every fixed deposit with no nominee recorded. Then show every fixed deposit whose nominee is not 'Priya Verma', **including** those with no nominee at all. Explain in one sentence why the second query needs more than a simple `<>`.

Q20. (Core) For each fixed deposit show the receipt number, principal, and a computed column `annual_interest` equal to principal times rate divided by 100, rounded to 2 decimals. Sort by annual interest descending and show the top five.

Q21. Produce a list for the branch, showing for every fixed deposit: receipt number, customer full name as one column, principal, and a `segment` column that reads HIGH VALUE at 1,000,000 or above, STANDARD at 100,000 or above, and SMALL otherwise. Sort high value first.

Q22. Find all transactions in `account_txn` whose narration came through the UPI channel and whose amount exceeds 50,000. Show transaction id, account id, amount and narration, largest first, limited to ten rows. Write the `LIMIT` clause three ways: PostgreSQL, SQL Server and Oracle.

Modify safely

Q23. Deposit 7009 is being broken early by the customer. Write the `SELECT` that identifies it, then the `UPDATE` that sets status to BROKEN and auto-renew to N, then the `SELECT` that verifies the change. Roll the whole thing back at the end.

Q24. (Core) The nightly maturity job marks as MATURED every ACTIVE deposit whose maturity date is on or before 30 June 2026. Write the `SELECT` first, note the count, run the `UPDATE`, and confirm the counts match. Roll back afterwards.

Q25. RD 8002 misses three instalments and is marked defaulted. Update `status` and leave `installments_paid` unchanged. Then, instead of deleting the row, close it the way a bank would, and explain in one line why.

Level 3 - Design and judgement

Q26. A vendor's data feed for the Smart Saver product supplies `principal_amount` as a text field, sometimes as "250000", sometimes as "2,50,000.00", and occasionally as "250000 INR". The vendor suggests you store the column as `VARCHAR(30)` and "parse it when needed". Write your response. Include what you would insist on, and where in the pipeline you would put the cleaning.

Q27. (Core) Two engineers disagree. One wants `ON DELETE CASCADE` on `fixed_deposit.customer_id`, arguing that closing a customer should tidy up their deposits automatically. The other wants `RESTRICT`. Take a position and defend it in five sentences, referring to what a bank actually does when a customer leaves.

Q28. ICICI Bank's `account_txn` table will hold 40 million rows. A new requirement says every transaction must record which employee authorised it, where applicable. Design the change: state the column, its type, its nullability, whether it gets a foreign key, and what you would do about the 40 million existing rows.

Q29. (Core) The `status` column on `fixed_deposit` is `VARCHAR(12)` with a `CHECK` restricting it to four values. An architect proposes replacing it with a separate `deposit_status` lookup table and a foreign key. List two advantages of each approach and say which you would choose for this table and why.

Q30. (Written, 8 to 12 lines.) A junior colleague has just run `UPDATE account SET status = 'DORMANT';` on the production database, with no `WHERE` clause and autocommit on. They come to you two minutes later. Write what you would do, in order, and then write the two changes you would make to the team's working practices so it does not happen again. Do not blame the individual.